

# BEVISIONEERS

THE MERCEDES-BENZ FELLOWSHIP

BioVil - Project Checkpoint 8  
**Prototyping Chronicles**



# Prototyping Chronicles: *Basic Monitoring Chart for Biodigester Management*

## "The Monitoring Chart Prototype: A Failed Attempt at Simplifying Biogas Management"

### What We Hoped to Achieve

Our objective was to create a basic monitoring chart prototype that would allow farmers to manually track feedstock inputs, biogas production, and other key parameters over time. We aimed to provide a simple tool that would help farmers visualize their biodigester's performance and make informed decisions regarding feedstock management.

### Why It Didn't Work

During the implementation of our monitoring chart prototype, we encountered several significant challenges:

1. **Inconsistent Usage:** He(farmer) often forgot to update the chart regularly or found it cumbersome to record data manually. This inconsistency resulted in false records, making it difficult to analyze trends.
2. **Lack of Real-Time Feedback:** The chart did not provide real-time monitoring capabilities. Without immediate feedback on biogas production or feedstock quality, he struggled to make timely adjustments to optimize their biodigester operations.
3. **Complexity of Data Interpretation:** He found it challenging to interpret the data collected on the chart. Without clear guidelines on how to analyze trends or understand the implications of their readings, he felt overwhelmed and uncertain about how to proceed.

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4. **Limited Engagement:** The monitoring chart lacked visual appeal and user-friendliness. Farmers expressed that the design was not engaging enough to encourage regular use, leading to abandonment of the tool altogether.

## Key Lessons Learned

- **Simplicity is Key:** A straightforward design is essential for user engagement. Tools must be easy to understand and use without requiring extensive training or technical knowledge.
- **Real-Time Monitoring is Crucial:** Incorporating real-time data collection features would significantly enhance the utility of monitoring tools. Farmers need immediate feedback to make informed decisions about their biodigester operations.
- **Educational Support is Necessary:** Providing training sessions on how to interpret data and utilize monitoring tools effectively is vital for ensuring that farmers can maximize their benefits.
- **Visual Appeal Matters:** Engaging designs can encourage regular usage. Future prototypes should prioritize aesthetics and usability to foster a positive user experience.

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## Refinement Process

To these challenges, we initiated a refinement process:

- We redesigned the monitoring chart with a more user-friendly layout, incorporating visual elements such as color-coded indicators for easier interpretation of data.
- We will explore to integrating simple digital tools that could automate data collection and provide real-time feedback on their phone, with visualizations.
- We will develop an educational program that included workshops on best practices for using monitoring tools and understanding biogas production dynamics.

## Learnings

This prototyping chronicle serves as a valuable reminder of the complexities involved in designing effective monitoring tools for biodigesters. By learning from our initial failures with the basic monitoring chart, we are committed to refining our approach and developing more effective solutions that empower farmers in managing their biodigester operations. This iterative process will ultimately contribute to improved biogas production and sustainability within communities.